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IB MATHEMATICS SL • UNIT 2.01

Topic 2: Functions, Transformations & Modeling

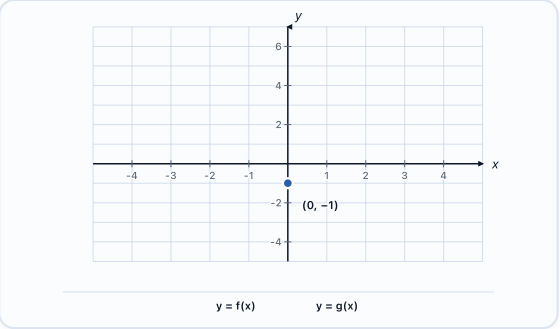
PRACTICE WORKSHEET

25 Structured Exam-Style Questions **Name:** _____ **Date:** _____ **Score:** _____ / 25

Topic 2: Functions, Transformations & Modeling Practice Exercises

1 MCQ

The graph of the function $f(x) = -x^2 + 4x - 3$ is shown below. What are the coordinates of the y-intercept?



A $(0, -3)$

B $(0, 3)$

C $(-1, 0)$

D $(3, 0)$

2 MCQ

Consider the function $f(x) = 2x + 3$. What is the value of $f(4)$?

A 7

B 11

C 10

D 5

3 MCQ

Consider the function $f(x) = 2x - 3$. Find the value of $f(3)$.

A 3

B 6

C 9

D 0

4 MCQ

Consider the function $f(x) = -x^2 + 6x - 5$. Determine the coordinates of the vertex of the graph of $y = f(x)$.

A (3, 4)

B (-3, 4)

C (3, -4)

D (4, 3)

5 MCQ

Consider the function $f(x) = 2x - 1$. Find the value of $f(3)$.

A 5

B 6

C 7

D 4

6 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown. Determine the value of $f(0)$.

A 1

B -1

C 3

D -3

7 MCQ

Consider the function $f(x) = -2x + 6$. The graph of $y = f(x)$ is shown below. Determine the value of the y-intercept.

A 6

B -3

C 3

D -6

8 MCQ

Consider the function $f(x) = x^2 - 4x + 3$. The graph of $y = f(x)$ is shown below. The function intersects the x-axis at two points. Find the coordinates of the point with the smaller x-value.

A (1, 0)

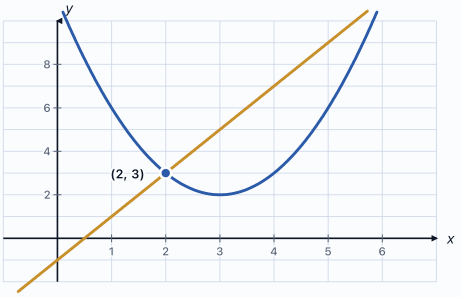
B (3, 0)

C (0, 3)

D (0, 1)

9 MCQ

Consider the functions $f(x) = x^2 - 6x + 11$ and $g(x) = 2x - 1$. The graphs of these functions intersect at two points. One of the intersection points is $(2, 3)$. Find the x-coordinate of the other intersection point.



A 3

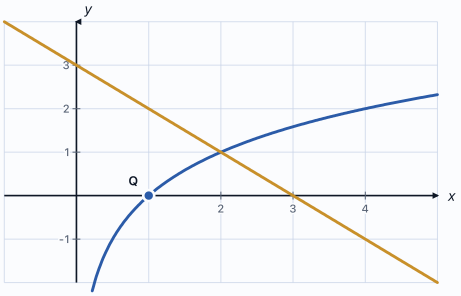
B 4

C 5

D 6

10 MCQ

Consider the two functions $f(x) = \log_2(x)$ and $g(x) = 3 - x$. The graphs of $f(x)$ and $g(x)$ intersect at a point P, and $f(x)$ intersects the x-axis at point Q. A line segment is drawn from Q to a point R on the graph of $g(x)$ such that QR is perpendicular to the x-axis. Find the coordinates of the point R.



A (4, -1)

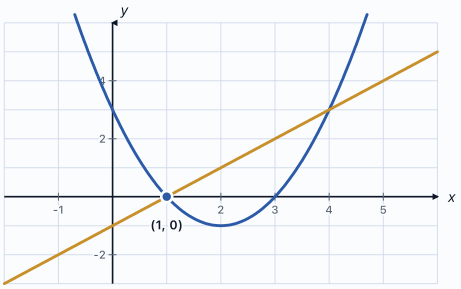
B (3, 0)

C (1, 2)

D (2, 1)

11 MCQ

Consider the function $f(x) = x^2 - 4x + 3$ and the line $g(x) = x - 1$. The graph of these functions is shown below. Find the coordinates of the point where the line intersects the parabola.



A (1, 0)

B (3, 2)

C (2, 1)

D (4, 3)

12 MCQ

Let $f(x) = 2x^2 + 6x - 5$. Find the y-intercept of the graph of f.

A -10

B 5

C -5

D -15

13 MCQ

The point $(0, -3)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 4$.

A $(1, -7)$

B $(-1, -7)$

C $(-4, -2)$

D $(1, 1)$

14 MCQ

In this problem, the point $(4, -9)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x - 1) - 2$.

A $(5, -7)$

B $(3, -11)$

C $(2, -8)$

D $(5, -11)$

15 MCQ

Let $f(x) = x^2 + 2x - 9$. Find $f(3)$.

A 6

B 2

C 3

D 1

16 MCQ

The point $(-4, 1)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(4, 1)$

B $(4, -1)$

C $(-4, -1)$

D $(-4, 1)$

17 MCQ

The point $(0, 7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(1, -7)$

B $(0, 7)$

C $(0, -6)$

D $(0, -7)$

18 MCQ

The point $(-1, -7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x + 4) - 1$.

A $(3, -8)$

B $(-5, -6)$

C $(-2, -11)$

D $(-5, -8)$

19 MCQ

The point $(6, -2)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(x + 5) + 5$.

A $(11, -7)$

B $(1, -7)$

C $(1, 3)$

D $(11, 3)$

20 MCQ

Let $f(x) = -x^2 + x - 6$. Find $f(2)$.

A -8

B -12

C -18

D 2

21 MCQ

Let $f(x) = -2x^2 + 8x - 2$. Find the y-intercept of the graph of f.

A 0

B 3

C -2

D 1

22 MCQ

Suppose we define $f(x) = -2x^2 + 7x - 1$. Find the y-intercept of the graph of f.

A 0

B -1

C 1

D 2

23 MCQ

The point $(-3, -7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = f(-x)$.

A $(3, 7)$

B $(-3, 7)$

C $(3, -7)$

D $(-3, -7)$

24 MCQ

In this problem, the point $(1, 7)$ lies on the graph of $y = f(x)$. Find the corresponding point on the graph of $y = -f(x)$.

A $(-1, 7)$

B $(1, 7)$

C $(-1, -7)$

D $(1, -7)$

25 MCQ

Let $f(x) = -x^2 - 7x - 4$. Find $f(-4)$.

A 8

B 3

C 4

D 5